

CHAPTER

1 A PRACTICAL INTRODUCTION TO MARTINI 3 AND ITS APPLICATION TO PROTEIN-LIGAND BINDING SIMULATIONS

Riccardo Alessandri, Sebastian Thallmair, Cristina Gil Herrero,
Raúl Mera-Adasme, Siewert J. Marrink, and Paulo C. T. Souza

Alessandri, R., Thallmair, S., Gil Herrero, C., Mera-Adasme, R., Marrink, S. J., and Souza, P. C. T., “A practical introduction to Martini 3 and its application to protein-ligand binding simulations,” in *A Practical Guide to Recent Advances in Multiscale Modeling and Simulation of Biomolecules*, edited by Y. Wang and R. Zhou (AIP Publishing, Melville, New York, 2023), pp. 1-1–1-34.

16 July 2024 15:18:01

1.1 INTRODUCTION

The Martini force field is one of the most popular coarse-grained (CG) models used in molecular dynamics (MD) simulations, with many applications in diverse fields, from structural biology (Abellón-Ruiz *et al.*, 2017; Yen *et al.*, 2018; and Liaci *et al.*, 2021) and biophysics (Agostino *et al.*, 2017; Vögele *et al.*, 2018; and Faustino *et al.*, 2020) to nanotechnology (Bruininks *et al.*, 2020; Zhang *et al.*, 2021; and Machado *et al.*, 2022) and materials design (Liu *et al.*, 2018; Vazquez-Salazar *et al.*, 2020; and De Marco *et al.*, 2021). One of the main characteristics of the Martini model is the use of a building-block approach. In this approach, small chemical fragments composed of two to four non-hydrogen atoms (but including associated hydrogens) are represented by CG particles, commonly called “beads.” The resulting simplification of the molecular systems is among the main reasons for the high computational performance of the model, which can be 2 to 3 orders of magnitude faster than atomistic force fields (see Note 1 for more details about Martini performance). Martini started to be developed around 2002 (Marrink and Mark, 2003a, 2003b), with the first version only used for the modeling of phospholipid systems (Marrink *et al.*, 2004). Over the years, the force field was expanded in a second version, Martini 2, which included new models of not only new and more specialized lipids (López *et al.*, 2013; Melo

et al., 2015; and *Gu et al.*, 2017), but also many other biomolecules such as proteins (*Monticelli et al.*, 2008; and *De Jong et al.*, 2013), carbohydrates (*López et al.*, 2009, 2015), nucleic acids (*Uusitalo et al.*, 2015, 2017), post-translational modifications (*Atsmon-Raz and Tieleman*, 2017; and *Shivgan et al.*, 2020), and a variety of cofactors and metabolites (*De Jong et al.*, 2015; and *Sousa et al.*, 2021). Martini 2 models also expanded towards materials science applications as models for polymers, surfaces, and nanoparticles were developed (*Alessandri et al.*, 2021). With the evolution of the model, limitations inherent in the coarse-graining approach (see Note 2) and also specific to Martini (see Note 3), in particular, the “stickiness” problem (*Alessandri et al.*, 2019), became clear. The necessary improvements to the model in combination with a high demand for better coverage of chemical space have promoted the development of the new version of the model called Martini 3 (*Souza et al.*, 2021a). In this chapter, we will focus on the main aspects of this new release of the Martini force field.

The Martini 3 model has a total of 7 chemical classes of beads: polar (P), intermediately polar (N), apolar (C), halo-compounds (X), monovalent ions (Q), divalent ions (D), and water (W) (Fig. 1.1). Except for W and D beads, all beads have subtypes that are distinguished by a number indicating the relative degree of polarity (from 1, low polarity, to a maximum of 6, high polarity). These subtypes allow for a more accurate representation of the chemical nature of the underlying atomistic structure. Furthermore, the beads can be distinguished by additional labels which mimic chemically specific interactions, such as hydrogen bonding (see Note 4 for details about other labels). All beads come in three different sizes, which are associated with the number of non-hydrogen atoms represented by the

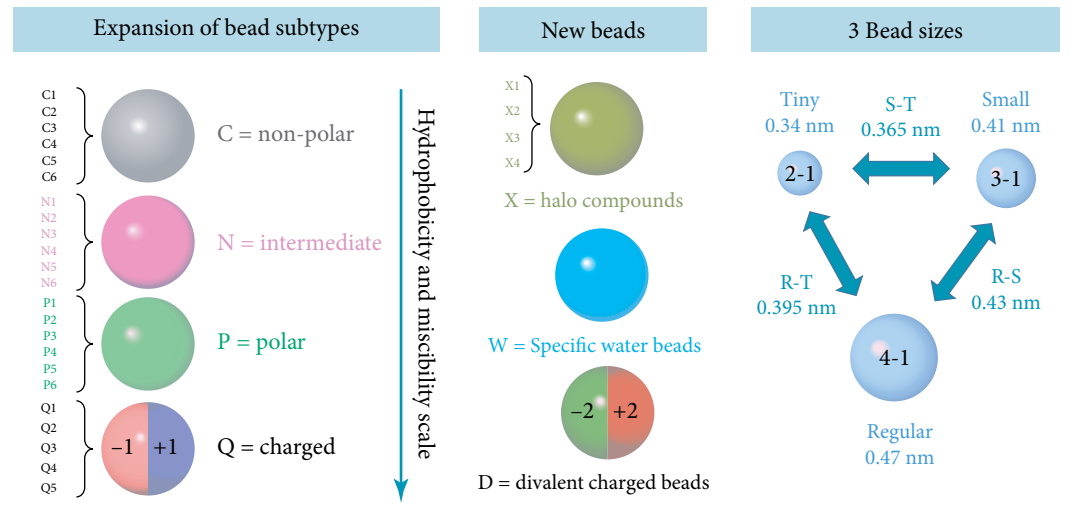


FIG. 1.1
Overview of bead chemical types and sizes in Martini 3.

bead: regular (“R,” for 4 non-hydrogen atoms represented by 1 bead, or 4-1), small (“S,” 3-1), and tiny (“T,” 2-1) (Fig. 1.1). These assignments are reasonable for linear fragments with exceptions such as in the case of branched moieties (see Note 5).

Martini 3 employs non-bonded and bonded potentials typical for atomistic force fields (and also in the previous Martini versions), which makes the model easily implementable in modern MD programs like GROMACS (Abraham *et al.*, 2015). Lennard–Jones (LJ) potentials are used to model the effective non-bonded interactions between all CG beads. Each pair interaction needs to be explicitly defined, with the interactions calibrated to reproduce thermodynamic data, such as liquid–liquid partitioning and miscibility. The sigma of the LJ potential defines the size of the beads, with a default size for the self-interaction of R, S, and T-beads of 0.47, 0.41, and 0.34 nm, respectively. Interactions involving special beads (such as W), cross-interactions between different sizes, and certain interactions between polar and non-polar beads may use different sigma values (Souza *et al.*, 2021a). In addition to LJ potentials, the charged beads (Q and D) use Coulomb interactions. These electrostatic interactions take into account a default explicit screening with a relative permittivity of $\epsilon_r = 15$. Both LJ and Coulomb potentials are shifted such that the potentials reach 0 kJ/mol for any distance greater than 1.1 nm, the cutoff distance (De Jong *et al.*, 2016).

While non-bonded interactions follow a “top-down” approach (making use of experimental data) for calibration, bonded interactions are typically obtained via a “bottom-up” approach based on reference atomistic MD simulations. Simple bonded potentials, such as two-body harmonic potentials, three-body angular potentials, and four-body periodic dihedral angle potentials, are commonly used in Martini. The parameters of these potentials are fitted according to the distributions obtained from atomistic simulations mapped onto a pseudo-CG trajectory. In contrast to previous versions, Martini 3 presents a more consistent set of rules for the calculation of bond lengths and also for the selection of bead sizes, both aspects aiming at an optimal representation of molecular volume and shape of CG models (Souza *et al.*, 2021a; and Alessandri *et al.*, 2022). The combination of improved packing of the CG models, better coverage of the chemical space, and improved interactions opened new application avenues for Martini. One such avenue is the study of protein-small molecule binding. We recently demonstrated that Martini 3 can achieve millisecond sampling and high accuracy of protein–ligand interaction (Souza *et al.*, 2020). Pockets and binding pathways can be predicted without the need of any *a priori* knowledge, with the approach being tested on a range of systems from the well-characterized T4 lysozyme to members of the GPCR, nuclear receptor, and kinase families (Souza *et al.*, 2020).

In Secs. 1.2–1.4, we will describe a hands-on tutorial illustrating the application of Martini to protein–ligand binding. The tutorial will use one of our protein–ligand binding studies as a guide: the binding of caffeine to the adenosine A2A receptor, which is part of the GPCR family. This tutorial will first illustrate with concrete examples of how to build Martini CG models of small molecules and transmembrane proteins. In the final part of the text, these CG models will be embedded in a water-lipid bilayer system to be used in unbiased protein–ligand binding simulations. Although this material is based on GROMACS (Abraham *et al.*, 2015) (versions 2018.X to 2020.X), similar approaches can be adapted

using other software packages that support Martini such as NAMD (Phillips *et al.*, 2020), LAMMPS (Jewett *et al.*, 2021), ddcMD (Zhang *et al.*, 2020), and openMM (Eastman *et al.*, 2017).

1.2 PARAMETERIZATION OF A NEW SMALL MOLECULE

In this section, we will describe how to create a Martini 3 model for a new small molecule—in this case, a model for caffeine. We will discuss considerations that are generally relevant for creating small molecules within the Martini 3 framework. As such, this section constitutes a pragmatic description of the Martini 3 coarse-graining principles described in Refs. (Souza *et al.*, 2021a) and (Alessandri *et al.*, 2022). The full procedure and necessary files are available at <https://github.com/ricalessandri/Martini3-small-molecules/>. Here, we will discuss in detail the procedure and highlight key steps and commands.

1.2.1 Generation of atomistic reference data

We need an all-atom (AA) reference simulation to extract the bonded parameters for the CG model. There is no particular constraint on the AA model to be used. Note only that, ideally, all the hydrogens are needed to extract optimal bonded parameters in Martini 3, so an all-atom—instead of a united-atom—model is the ideal choice.

In this case, we use an OPLS-AA model as obtained from the LigParGen server (Jorgensen and Tirado-Rives, 2005; and Dodda *et al.*, 2017a, 2017b). The model can be obtained from the server by providing the SMILES string of caffeine (i.e., Cn1cnc2n(C)c(=O)n(C)c(=O)c12). LigParGen's OPLS-AA/1.14*CM1A-LBCC files in the GROMACS format for caffeine can also be found on the Martini 3 small molecule database (Alessandri *et al.*, 2022). We solvate the caffeine AA model in TIP3P water (Jorgensen *et al.*, 1998) and after energy minimizing and equilibrating the system we run an MD simulation of 50 ns. Note that caffeine is a fairly rigid small molecule; hence, this simulation time is more than sufficient to sample all the bonded degrees of freedom. In the case of more complex small molecules, a longer simulation time may be required for adequate sampling. Generating the atomistic reference data requires running a standard MD simulation with GROMACS (Abraham *et al.*, 2015). The files necessary for the run along with a script with all the commands can be found at the GitHub repository: <https://github.com/ricalessandri/Martini3-small-molecules/>.

1.2.2 Definition of the coarse-grained mapping

Splitting the molecules in building blocks to be described by CG beads, i.e., CG mapping, is a cornerstone of coarse-graining. Defining a CG mapping within Martini 3 should follow the guidelines below. Moreover, the use of already available models (Alessandri *et al.*, 2022) as a reference and starting point is another useful practice to develop Martini 3 mappings in a rapid and more consistent way.

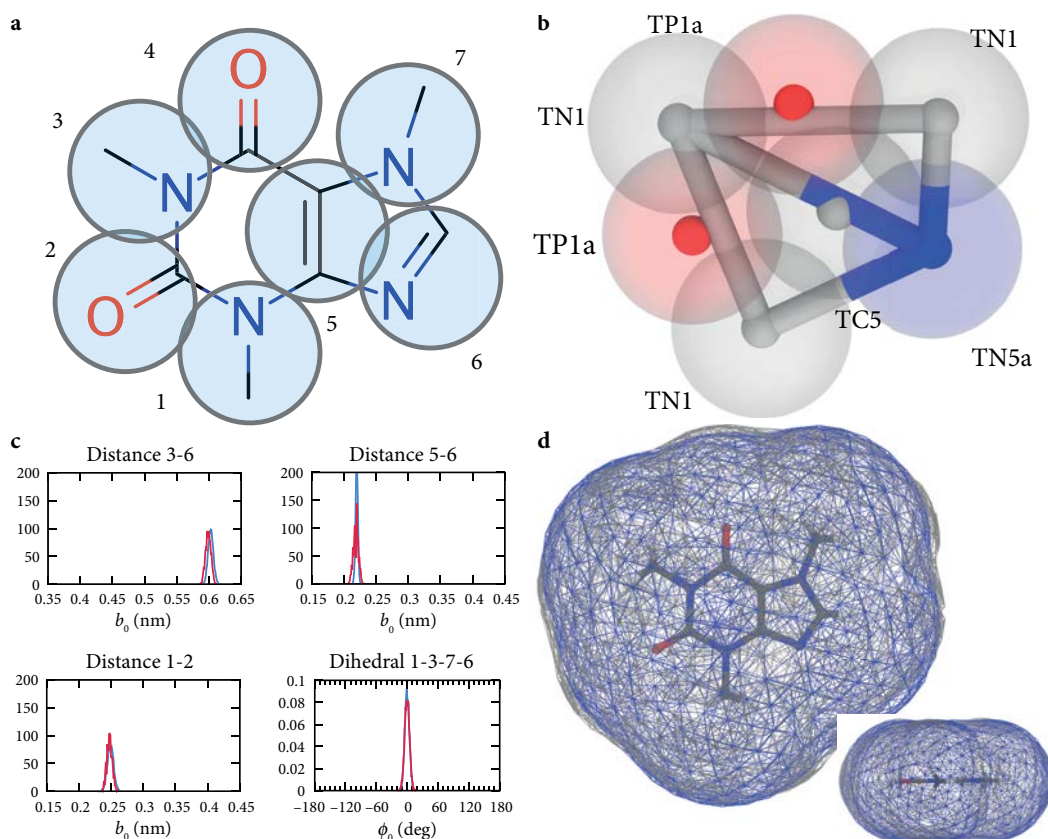
- Consider only non-hydrogen atoms to define the mapping.
- Avoid dividing chemical functional groups (e.g., amide, carboxylate, etc.) between two beads.
- Retain in the CG representation as much as possible (1) the *symmetry* and (2) the *molecular volume and shape* of the underlying AA structure.
- Keep in mind that the default bead sizes for mapping linear molecular fragments are one regular (R) bead for 4 non-hydrogen atoms (4-1), one small (S) bead for three (3-1), and one tiny (T) bead for two (2-1).
- Consider that R-beads are the most computationally-performant option; S-beads are useful to mimic the “bulkier” shape of aliphatic rings, while T-beads are particularly suited to represent atom-thick, conjugated rings.
- Optimize the number of beads such that there is at most a mismatch of ± 1 non-hydrogen atom every 10 non-hydrogen atoms. Some exceptions may be allowed if well-tested (e.g., mapping of thiophene or furan with three T-beads have a mismatch of -1 given that 5 non-hydrogen atoms are mapped onto 3 T-beads which are parameterized to represent 6 non-hydrogen atoms).
- For fully branched fragments, use beads of one size smaller than the size determined based on the non-hydrogen atom count. The rationale behind this is that the central atom of a branched group is not exposed to the environment (i.e., it is buried), which reduces its influence on the interactions. For example, a neopentane group contains 5 non-hydrogen atoms but it can be effectively modeled with an R-bead (see also Note 5).

In the present case, as the structure of caffeine is mostly fully conjugated, T-beads are a good choice to ensure accurate packing-related properties, which are important for ligand binding recognition (Souza *et al.*, 2020, 2021a; and Alessandri *et al.*, 2022). Given a non-hydrogen atom count of 14, caffeine can be mapped to 7 T-beads, as shown in Fig. 1.2(a), where functional groups—such as carbonyl groups—have been intentionally kept together.

1.2.3 Coarse-grained mapping of the atomistic reference data

Using the mapping just defined [Fig. 1.2(a)], we now need to transform the AA trajectory to CG resolution. In order to do this, we create a GROMACS (AA-to-CG) index file where every index group stands for a CG bead and contains the corresponding mapped atom numbers.

The required index file can be generated in several ways. One way which may be useful to define and refine a mapping is to use the CGbuilder tool (CGbuilder). CGbuilder allows previewing and refining mappings via an intuitive GUI. The GUI requires the AA structure file (e.g., `pdb` or `gro` format) and allows to define the mapping by clicking on the atoms to add/remove them to/from a certain bead. A current limitation of CGbuilder is that atoms cannot contribute with a weight different from one to a certain bead, something which is at times necessary to keep the symmetry of the underlying AA structure in its Martini representation. In such cases, the index file must be subsequently refined by editing the file created by CGbuilder.

**FIG. 1.2**

Martini 3 CG model of caffeine. (a) The 14 non-hydrogen atoms are described by a 7 T-bead model; the indices used for the beads in the CG topology file are also shown. (b) Rendering of the CG model: apolar aromatic and intermediately polar beads are displayed in silver (TC5, TN1) and blue (TN5a) while polar (TP1a) beads are in red. As described in Sec. 2.5 and shown in the rendering, beads 1, 3, 6, and 7 are connected via constraints and form a “hinge” construction, while beads 2, 4, and 5 are constructed as virtual sites. (c) Representative bond and dihedral distributions: OPLS is in blue while Martini is in red. Note that while distance 3-6 corresponds to an actual constraint at the CG level, distances 5-6 and 1-2 at the CG-level result from the virtual site constructions. (d) A comparison of the Connolly surfaces of the AA (gray) and Martini 3 (blue) models; the inset shows a side view of the molecule.

An important aspect to take into account when creating the CG mapping file is that in Martini 3 *hydrogen atoms are taken into account to determine the relative position of the beads*, in what we refer to as center-of-geometry (COG)-based mapping (Souza *et al.*, 2021a; and Alessandri *et al.*, 2022). This aspect constitutes a change with respect to the previous versions of Martini, which instead neglected

the hydrogens also when deriving bonded parameters. COG-based mapping leads to Martini 3 models with better molecular (e.g., solvent accessible surface area) and bulk (e.g., mass densities) properties (Souza *et al.*, 2021a; and Alessandri *et al.*, 2022). Hence, COG-based mapping of AA structures taking into account the hydrogen atoms is the standard procedure for deriving bonded parameters in Martini 3. Accordingly, in the case of caffeine, the first two groups of the index file representing the first two beads will look as follows:

```
[ B0 ]
19 7 21 20 6
[ B1 ]
9 8,
```

where B0 indicates the first bead and it maps 5 atoms—the N atom and the four atoms of the methyl group, as shown in Fig. 1.2(a); the second bead B1 instead describes only 2 atoms, the C and O of the carbonyl group [Fig. 1.2(a)].

With the index file in place, we can now convert the AA trajectory to CG resolution according to our mapping. First, we make the AA trajectory “whole”:

```
echo 0 | gmx trjconv -f ../1_AA-reference/3-run.xtc -o AA-
traj.whole.xtc -s ../1_AA-reference/3-run.tpr -pbc whole
```

We then create a COG-based topology file of the AA system (see Note 6):

```
gmx grompp -p system_COG.top -f ../1_AA-reference/run.mdp -c
../1_AA-reference/2-eq.gro -o AA-COG.tpr
```

and then map the trajectory with the GROMACS tool `gmx traj`:

```
seq 0 6 | gmx traj -f AA-traj.whole.xtc -s AA-COG.tpr -oxt
mapped.xtc -n mapping.ndx -ng 7 -com,
```

which will produce the CG-mapped trajectory `mapped.xtc`.

1.2.4 Generation of the initial coarse-grained model

To define the CG model of caffeine, a CG topology file needs to be generated. In GROMACS, such a topology is defined by creating an `itp` file. This file contains the definition of the CG bead types and properties as well as the bonded parameters that define how the beads are held together. A thorough description of the `itp` file format can be found in the GROMACS manual (GROMACS 2021.5). Here, we recap the basics and describe the terms necessary to create a CG model for caffeine.

The first entry in the `itp` file is the `[moleculetype]`, which contains the molecule name and the number of non-bonded exclusions. In Martini, the standard number of exclusions is 1, meaning that

non-bonded interactions between particles directly connected are excluded. Hence, in the case of caffeine, the first lines read

```
[moleculetype]
; molname      nrexcl .
  CAFF          1
```

However, for models of rigid rings, it is common to exclude all the intramolecular non-bonded interactions. In particular, virtual sites need to be excluded in relation to all neighboring real beads. To do that, we explicitly add an [exclusions] section, typically at the end of the `itp` file. Each line of this section indicates that the bead in the first column does not interact with the beads in the next columns of the same line:

```
[exclusions]
1 2 3 4 5 6 7
2 3 4 5 6 7
3 4 5 6 7
4 5 6 7
5 6 7
6 7
```

The next entry after [moleculetype] is [atoms], where all the beads that make up the molecule are defined. For each bead, we define a one-line entry containing *bead number*, *bead type*, *residue number*, *residue name*, *bead name*, *charge group*, *charge*, and *mass*. Bead numbers should start from one. In the case of small molecules like caffeine, the residue number and residue name will be the same for all beads. Bead names can be freely chosen. The charge and charge group are not relevant in the case of caffeine given its charge neutrality. The mass and bead type assignments will be described in more detail below.

```
[atoms]
; nr type resnr residue atom cgnr charge mass
  1 TN1    1    CAFF   R1    1    0    63
  2 TP1a   1    CAFF   O2    2    0    0
  3 TN1    1    CAFF   R3    3    0    63
  4 TP1a   1    CAFF   O4    4    0    0
  5 TC5    1    CAFF   R5    5    0    0
  6 TN5a   1    CAFF   N6    6    0    63
  7 TN1    1    CAFF   R7    7    0    63
```

1.2.4.1 Bead types

Similar to atom types in AA force fields, in Martini, we must assign bead types to the different beads that compose a given molecule. The initial assignment can be done based on the underlying chemical building block, following the examples collected in Table 1.1 of Alessandri *et al.*, 2022 [note that an up-to-date version of the table can be found at (<https://github.com/ricalessandri/Martini3-small-molecules/blob/>]

Table 1.1

Computed and experimental free energies of transfer for caffeine. Free energies of transfer between (hydrated) octanol (OCO), hexadecane (HD), chloroform (CLF), and water (W) are reported. Values are in kJ mol⁻¹. Statistical uncertainty for the computed values is below 0.3 kJ mol⁻¹ in all cases.

Model	$\Delta G_{OCO \rightarrow W}$		$\Delta G_{HD \rightarrow W}$		$\Delta G_{CLF \rightarrow W}$	
	Expt.	CG	Expt.	CG	Expt.	CG
Take 1: wrong bead assignment TN4a-TN1-TN4a-TN1-TC5-TN6a-TN1	-0.4	5.8	...	-12	...	6.2
Take 2: based on the reference table TN6a-TN1-TN6a-TN1-TC5-TN6a-TN1	-0.4	1.2	...	-17.9	...	-2.2
Take 3: Analogy to nucleobases TP1a-TN1-TP1a-TN1-TC5-TN5a-TN1	-0.4	1.6	...	-19.1	...	1.0

[main/tutorials/building_block_table.pdf](#)]). It is important to note that, after the initial assignment, bead type choices can be subsequently fine-tuned based on, for example, comparison to experimental free energy of transfer data. This is the case for the beads TP1a, which represent carbonyl groups (see Sec. 2.8). The default assignment would be TN6a in the case of simple ketones and aldehyde groups. However, as the carbonyl groups in caffeine are conjugated in the aromatic heterocycle, their bead type can be subject to a deviation from the standard assignment as it was considered for nucleobases (Souza *et al.*, 2021a).

1.2.4.2 Masses

When masses are not specified—as is usually the case in Martini—default masses of 72, 54, and 36 a.m.u. are used for R-, S-, and T-beads, respectively. However, when virtual sites are present, such as in the present case, masses should be specified to account for the mass-less virtual sites (see Sec. 2.5). In these cases, a pragmatic choice is to redistribute the mass of the whole molecule between the real beads (which is our option here). An alternative would be to use the real masses of the fragment mapped in the real beads combined with the real masses of the fragment mapped as a virtual site, possibly only redistributing the masses between the real beads used in the construction of the virtual site position.

The [moleculetype] and [atoms] entries are mandatory and make up a basic itp file. With these entries, we can generate a Gromacs binary topology file, CG.tpr, that will be necessary later on. To do this, one needs the basic topology file just defined for the CG model (system_CG.top), a file containing the simulation settings (which can be found in folder 4_initial-CG; let's say we pick martini_v2.x_new_run.mdp, although this is not important at this stage), and a coordinate file. The latter can be extracted from the atomistic trajectory in the following way:

```
seq 0 6 | gmx traj -f ../3_mapped/AA-traj.whole.xtc -s
../3_mapped/AA-COG.tpr -n ../3_mapped/mapping.ndx -oxt
molecule.gro ng 7 -com -e 0
```

16 July 2024 15:18:01

that the `CG.tpr` can then be generated with the following command:

```
gmx grompp -f martini_v2.x_new_run.mdp -c molecule.gro -p
system_CG.top -o CG.tpr -maxwarn 1
```

Further necessary entries in order to define a multiparticle model of a molecule are entries defining bonded terms such as `[bonds]`, `[constraints]`, `[dihedrals]`, etc. We will go through these terms in Sec. 1.2.5.

1.2.5 Generation of bonded terms and target distributions

Next, parameters for the bonded interactions need to be obtained from the AA trajectory mapped to the CG resolution. In this subsection, we first describe how to settle on which bonded parameters we need, and then translate that into index files that are needed in order to extract the reference distributions from the CG-mapped trajectory.

1.2.5.1 The choice of bonded terms for the coarse-grained model

The bonds connecting the beads within the CG model of caffeine are expectedly very stiff, i.e., their distributions are very narrow. At the Martini scale, the fast stretching of these bonds becomes unimportant, and, hence, they can be replaced by constraints, i.e., those bond lengths are held fixed (Marrink *et al.*, 2007; Souza *et al.*, 2021a; and Alessandri *et al.*, 2022). A straightforward but naïve way of constructing the model would be to interconnect all the beads with constraints. However, such a model would easily lead to numerical instabilities, as the constraint solver—we assume here the use of the LINCS (Hess *et al.*, 1997) algorithm in GROMACS—cannot satisfy a growing number of connected constraints with a time step of 20 fs.

A way to circumvent the use of more than five interconnected constraints and, hence, build a numerically stable model is to use a so-called “hinge” construction (Melo *et al.*, 2015; and Alessandri *et al.*, 2022). In such construction, four beads are used to build a hinge, while other beads are constructed as virtual sites, i.e., particles the position of which is defined by their constructing particles. The use of virtual particles not only improves the models’ numerical stabilities but also their performance (Alessandri *et al.*, 2022). In the specific case of caffeine, beads number 1, 3, 6, and 7 can be connected via 5 constraints to construct the hinge, as shown in Fig. 1.2(b). This means having one such line in the `itp`:

```
[constraints]
; i j funct length
1 3 1 XXX
```

for each constraint to be defined, namely constraints 1-3, 1-6, 3-6, 3-7, and 6-7. Here, XXX is a placeholder for the initial constraint length that will be obtained from the target atomistic distribution in the next step. An improper dihedral potential involving the 4 constructing beads needs to be added

to the model to keep the beads in the same plane (e.g., dihedral 1-3-7-6 with an equilibrium value set to zero). The remaining three beads can be defined via virtual sites as follows. First, bead number 5 can be defined with GROMACS' `[virtual_sites2]`, that is, its position can be constructed as a linear combination of the positions of beads 3 and 6:

```
[virtual_sites2]
; sites positioned at 1-a
; site from funct a
  5    3 6    1    0.635
```

The factor *a* specifies the position of bead 5 on the connecting line between beads 3 and 6 ([GROMACS 2021.5](#)). The remaining two beads, bead number 2 and 4 [Fig. 1.2(b)], can be defined using the `[virtual_sites3]` construction as follows:

```
[virtual_sites3]
Site from      funct a      b
  2    5 1 3      1    0.640 0.605
  4    5 3 7      1    0.650 0.600
```

In such construction, the site is positioned as a linear combination of the position of three particles (e.g., in the case of bead 2, these are beads 5, 1, and 3). Note that both `[virtual_sites3]` constructions use bead 5 as a constructing particle, which in turn is constructed as a virtual site itself. This dependency is possible because bead 5 is a “simpler” virtual site, i.e., it is constructed using a simpler construction than the construction in which it is used as constructing particle. Again, factors *a* and *b* specify the exact position of the virtual sites. More details on the virtual sites available in GROMACS can be found in the GROMACS manual ([GROMACS 2021.5](#)).

1.2.5.2 Index files and generation of the target distributions

Now that we have settled on the bonded terms, we can create index files for the bonds, angles, and dihedrals for which we want to plot the target distributions. For example, the file `bonds.ndx` will contain an entry `[bondX]` for each bond, each containing pairs of CG beads between which a bond (or constraint) exists:

```
[bond1]
  1    3
[bond2]
  1    6
...
```

and similarly for the dihedrals (with quartets of CG beads); note that, in general, an `angles.ndx` (with triples) would have to be created for the angles, but in the case of caffeine no angle potentials are

needed. The following commands will then generate a bond length distribution for the first bond in the index (the bond connecting beads 1 and 3):

```
echo 0 | gmx distance -f ../3_mapped/mapped.xtc -n bonds.ndx
-s ../4_initial-CG/CG.tpr -oall bond_0.xvg -xvg none

gmx analyze -f bond_0.xvg -dist distr_bond_0.xvg -xvg none -
bw 0.001
```

similarly, for the first (and only) dihedral, we can issue the following commands:

```
echo 0 | gmx angle -type dihedral -f../3_mapped/mapped.xtc
-n dihedrals.ndx -ov dih_0.xvg

gmx analyze -f dih_0.xvg -dist distr_dih_0.xvg -xvg none -bw
1.0
```

1.2.6 Optimization of the coarse-grained parameters

We can now finalize the CG model by running a CG MD simulation of the caffeine CG model solvated in water after having energy-minimized, and equilibrated the system. The starting values for the bonded parameters can be obtained from the mean values of the distributions in the previous step. Further refinement of the bonded parameters of the CG model will be done by extracting bonded distributions from the CG simulations—using commands similar to those given in Sec. 2.5.2—and comparing them to the CG-mapped AA target distributions obtained in Sec. 1.2.5.

Representative distributions of well-fitted bonded parameters are shown in Fig. 1.2(c), where Martini distributions are shown in red and CG-mapped OPLS-AA in blue. It should be mentioned that to fine-tune the parameters for the virtual site constructions, one usually needs to compare distances between beads which are not explicitly connected via bonded potentials in the CG model. This is the case, for example, of the distributions 1-2 and 5-6 of [Fig. 1.2(c)].

1.2.7 Bartender: automated generation of coarse-grained bonded parameters

Bartender is a program for the automated determination of bonded parameters for Martini 3 ([Bartender](#)). As such, its use can replace the steps in Secs. 1.2.1, 1.2.3, and 1.2.6. Bartender takes advantage of the novel “GFN” semiempirical quantum-chemical methods developed by the Grimme group in recent years ([Bannwarth et al., 2021](#)). It employs one of these methods to perform a Born–Oppenheimer quantum mechanical MD (QM-MD) simulation. Subsequently, the bonded parameters are fitted to the distance, angle, and dihedral distributions from the QM-MD trajectory. The GFN methods are efficient enough to run dynamics for several nanoseconds, allowing a good sampling of the conformational space for most rigid small molecules like caffeine. For more challenging cases, Bartender includes replica exchange MD capabilities. In addition, Bartender can read a supplied trajectory in `dcd` or `xtc` format and omit the QM-MD step.

1.2.7.1 Input files

Bartender requires two input files. The first one is a structure file in `pdb`, `gro`, or `xyz` format with the atomistic representation of the molecule. The second input file, which is specific to Bartender, contains a description of the desired atom-to-bead mapping, and of the bonded parameters that are to be obtained. The file is divided into several sections separated by headers. This input file can be created using the python script `write_bartender_inp.py` included in the Bartender GitHub repository (<https://github.com/rmera/bartender>). A sample input for the caffeine molecule is given below:

```
BEADS
1 19,6,21,720
2 8,9
3 23,24,11,22,10
4 12,13
5 514
6 18,3,4
7 15,1,17,16,2
BONDS
1,3
1,6
3,6
3,7
6,7
IMPROPERS
1,3,6,7
```

The first section starts with the header `BEADS`. It contains a list of the CG beads in the Martini representation of the molecule, followed by a space, and a list of the indexes for the atoms that the bead represents, separated by a comma. A simple line for the section would be:

```
2 8,9
```

which indicates that bead 2 represents atoms 8 and 9. It is possible that an atom needs to be split between beads. That can happen for various reasons, such as to preserve the symmetry of the underlying atomistic structure. Bartender allows to define a bead containing only a fraction of an atom. For instance, the following lines:

```
2 8/2,9
3 8/2,23,24,11,22,10
```

divide atom 8 equally between beads 2 and 3. The other sections of the inputs are all optional and define the particular bonded terms that are to be obtained by Bartender. Their headings are `BONDS`, `ANGLES`, `DIHEDRALS`, and `IMPROPERS`. Each term is described in its own line by simply specifying the relevant

beads. For caffeine, no angles and dihedrals are necessary, only bonds and improper dihedrals. Virtual sites are still not implemented in Bartender, but they will be in the near future.

1.2.7.2 Running Bartender

When called, Bartender will perform a QM-MD for the given molecule and fit the requested bonded terms to their respective probability distributions in the obtained trajectory, using the Boltzmann distribution to convert frequencies into potential energies.

Bartender is a command-line program that can be executed as follows:

```
bartender [flags] geometry.xyz/pdb/gro bartender_input.inp
```

Omitting any flags would execute Bartender's default behavior, which is designed to nicely perform its purpose in most cases. However, the settings of Bartender can be modified using the available flags. The list of all available flags is detailed by calling Bartender as follows:

```
bartender -help
```

Some of the common flags are

<code>-charge <i>int</i></code>	Sets the total charge of the molecule (default: 0)
<code>-method <i>string</i></code>	Sets the specific GFN method to be used (default: GFNFF)
<code>-dcdSave <i>file.dcd</i></code>	Saves the QM-MD trajectory in DCD format.
<code>-owntraj <i>file.dcd/xtc</i></code>	Uses an MD trajectory instead of performing a QM-MD simulation.
<code>-time <i>int</i></code>	The total QM-MD simulation time in ps (default: 1000)
<code>-solvent <i>string</i></code>	Sets the implicit solvent to be used (default: h2o, for water)
<code>-cpus <i>int</i></code>	The number of CPUs used for the QM-MD simulation.

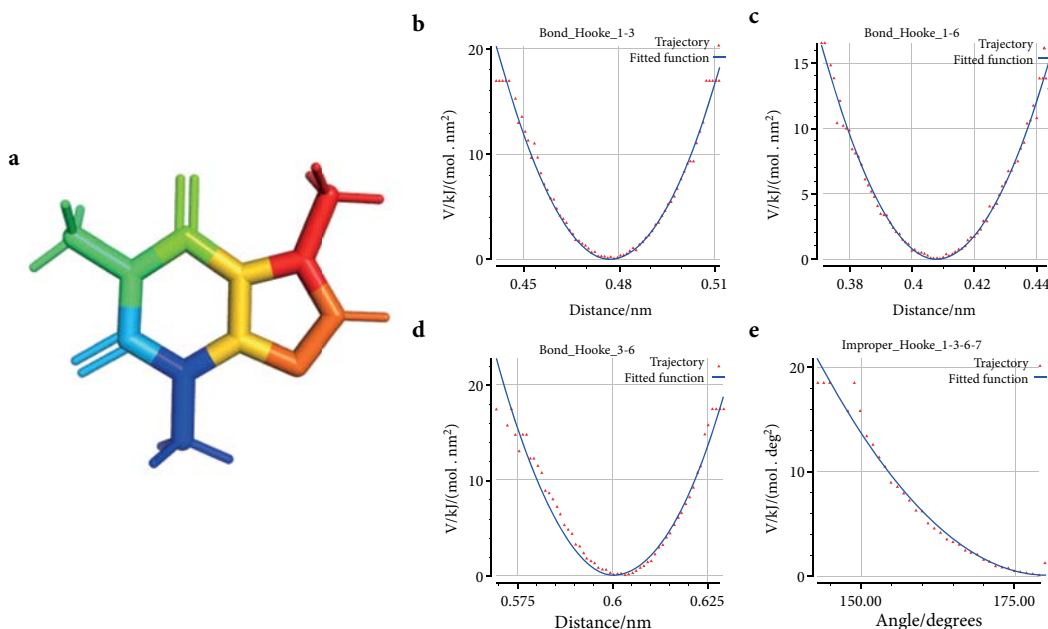
For our example, we need to generate a `xyz` file from the LigParGen `pdb` file, as LigParGen produces files that do not follow the `pdb` standard and cannot be read by `goChem`. The conversion can be done either by hand with custom scripts or by using programs such as `openbabel` or `VMD`. Bartender is then executed as follows:

```
bartender CAFF.xyz caff_bartender.inp
```

1.2.7.3 Bartender output

The Bartender run produces several output files:

1. A GROMACS topology file, `gmx_out.itp`, with the parameterization of the requested bonded terms.

**FIG. 1.3**

Examples of outputs from Bartender. (a) The `beads.pdb` file for the caffeine molecule, colored by b-factors. Atoms belonging to the same bead appear grouped by color. (b)–(e) Energies from the trajectory frequency distributions (red triangles) and from the fitted function (blue lines) for some of the bonded terms in caffeine.

2. A series of `png` files, each of which contains the plot of a fitted function for a bonded term, superimposed with the potential obtained from the Boltzmann inversion of distribution for that term in the MD.
3. A `pdb` file of the atomistic system, where the residue identifier for each atom contains the ID of the corresponding bead. The b-factor column of the `pdb` file is written in such a way as to group the atoms in a given bead by color when colored by b-factor, which is supported by viewers like PyMOL (Schrödinger, 2015) and VMD (Humphrey *et al.*, 1996) [see Fig. 1.3(a)].

The `gmx_out.itp` file is the most important one. For each requested bonded term, a corresponding line is written containing the parameters from the QM-MD fit. Some of the bonded terms can be parameterized with several different functions in GROMACS. Of these, Bartender supports different functions for angle (bending) and dihedral terms. Parameters are determined by fitting all available functions for each term. All fits are then printed to the `gmx_out.itp` file. For each term, all but one of the available fits are commented out so as to be ignored by GROMACS. To choose the function to be used for each term, Bartender prints the RMSD for each fit to the `gmx_out.itp` file. In addition, for

each fit, a PNG file is produced, which allows comparing the fitted function to the actual distribution from the QM-MD trajectory. In this study, only bonds and one improper dihedral are needed, and Bartender only supports harmonic potentials for them. Of course, it is possible to express the improper dihedral term as a regular dihedral, which would allow other functions to be tested.

In Figs. 1.3(b)–1.3(e), plots for some of the bonded terms are shown. Note that, for the three bonds shown, the fitting is generally good, though anharmonicities at the extreme points result in slight deviations of the fits. For the improper dihedral term, the fitting is also good except for an outlier close to 180 degrees. Because of the form of the potential, Bartender forces the equilibrium angle to 180 degrees, which appears to be a good approximation in this case. To account for deviations from planarity in the QM-MD simulation, a regular dihedral function could be employed alternatively. To finalize the caffeine model generated by Bartender, it is now necessary to add the constructions of the three virtual sites (see Sec. 1.2.5).

1.2.8 Validation and refinement of the coarse-grained model

We now describe how the model can be validated and refined and discuss some final considerations about creating models for small molecules with Martini 3.

1.2.8.1 Free energies of transfer

Experimental free energies of transfer are particularly important reference data for validating Martini CG models. In the case of caffeine, an experimental $\log P$ value is available for comparison (Hansch *et al.*, 1995). Table 1.1 reports a comparison between this value and the computed free energy of transfer of three different models.

The first take represents the results obtained with a wrong bead assignment for the carbonyl groups. In take 2, we follow the reference table in (Alessandri *et al.*, 2022) for the bead assignments and, hence, represent the carbonyl groups with TN6a beads. This modification is enough to get an improved prediction of the octanol-water free energy of transfer. In take 3, the model is changed considering its analogy with other nucleobases. The model still gets a reasonable octanol-water free energy of transfer, with some improvement of the strength and selectivity of hydrogen bonds with other nucleobases (Souza *et al.*, 2021a). Both take 2 and 3 show good agreement between the Martini model and experiments, with an error ≤ 2 kJ mol⁻¹. We note that Martini free energy values within $k_B T$ (2.479 kJ mol⁻¹) are considered to be in very good agreement.

1.2.8.2 Molecular volume and shape

Looking into how well the Martini CG model represents the underlying atomistic molecular size and shape is a way to further refine the model. It is also good practice to validate the size and shape of the CG model whenever molecular packing is critical for certain applications, which is the case for the ligand-protein interactions described in this chapter. In fact, the use of COG-based bonded parameters is oriented to high-throughput applications where this procedure could be automated. However, such

parameters may need refinements in some cases. In case the packing and/or mass density of a certain model is off, it is worthwhile to compare the molecular volume and shape of the CG model to the ones of the corresponding AA structure.

To this end, the GROMACS tool `gmx sasa` can be used to compute the solvent accessible surface area (SASA) and the Connolly surface for both the AA and CG models. In order to use this tool for the CG models, an appropriate `vdwradii.dat`, a file that contains the radius of atoms or beads and that is used by `gmx sasa`, needs to be provided. According to the naming provided in Sec. 1.2.4, the `vdwradii.dat` for the CG model will look like the following:

```
??? R1    0.191
??? O2    0.191
??? R3    0.191
??? O4    0.191
??? R5    0.191
??? N6    0.191
??? R7    0.191
```

Note that the file contains the *atom names* of the beads (second column), which in the case of caffeine, should be all set to the T-bead radius, 0.191 nm. R- and S-beads have a radius of 0.264 and 0.230 nm, respectively. Here, ??? in the first column indicate that any *residue* or *molecule name* could match with the *atom names* defined here. The user could use CAFF in case there are more molecules or residues to be defined in the `vdwradii.dat` file. This file should be copied to the location where the `gmx sasa` analysis is executed. We also recommend using more recent van der Waals radii ([Rowland and Taylor, 1996](#)) for the atomistic reference SASA calculations instead of the GROMACS default values given in the standard `vdwradii.dat` file.

Now, one can compute the AA SASA with the following command:

```
echo CAFF | gmx sasa -f ../3_mapped/AA-traj.whole.xtc -s
../3_mapped/AA-COG.tpr -ndots 4800 -probe 0.191 -o SASA-
AA.xvg
```

and similarly for the CG SASA. Note that the probe size is the size of a T-bead (the size does not affect the results particularly, as long as one compares SASA values obtained with the same probe size, naturally). The `-ndots 4800` guarantees a high accuracy of the computed SASA value. For caffeine, we obtain an AA SASA value of $4.37 \pm 0.01 \text{ nm}^2$ and a CG SASA value of $4.34 \pm 0.01 \text{ nm}^2$, which are in excellent agreement (discrepancy of <1%).

One can further compare how the shape of the CG model compares to that of the AA model by calculating the Connolly surfaces of the models. This can also be done with the `gmx sasa` tool as follows (AA case):

```
echo CAFF | gmx sasa -f AA.gro -s ../3_mapped/AA-COG.tpr -o
temp.xvg -probe 0.191 -ndots 240 -q surf-AA.pdb
```

Note that in this case, a lower accuracy (`-ndots 240`) is used to compute the Connolly surfaces (file `surf-AA.pdb`) in order to avoid excessively large surface files that can make visualization unnecessarily slow. The AA and CG surfaces can be visualized using VMD ([Humphrey *et al.*, 1996](#)) with the following command:

```
vmd -m AA.gro surf-AA.pdb surf-CG.pdb
```

Snapshots are displayed in Fig. 1.2(d), which show that the CG model faithfully reproduces the molecular shape and volume of the underlying AA representation, in line with the obtained SASA values. Given the small differences in bond lengths obtained using Bartender, it is also expected that SASA would be in good agreement in this case.

1.3 BUILDING A PROTEIN MODEL IN MARTINI 3

After successfully generating a CG model for the ligand caffeine, in this section, we will go through the steps required to generate a Martini 3 protein model. In Martini 3, the protein models contain two layers: the first layer contains the regular bonded terms of the protein backbone as well as the amino acid side chains and the second layer consists of the structure-preserving model typically required for proteins in a CG representation to maintain their higher-order structure ([Kmiciek *et al.*, 2016](#)). It is important to note that in Martini 3, the two layers are designed in such a way that they are independent of each other. This is in contrast to Martini 2, where different backbone mappings were applied for the regular Martini 2 elastic network (EN) approach ([Monticelli *et al.*, 2008](#)), the ElNeDyn (elastic network dynamics) approach ([Periole *et al.*, 2009](#)), and the GōMartini approach ([Poma *et al.*, 2017](#)). In this section, we will describe two options for the second layer, namely an EN model and a Gō-like model.

1.3.1 Preparation of the atomistic protein structure

We use an experimental X-ray crystal structure as a starting point to construct the Martini 3 protein model of the adenosine A2A receptor. First, download the structure with the `pdb` code 3RFM ([Doré *et al.*, 2011](#)) from the protein data bank (PDB).

```
wget https://www.rcsb.org/pdb/files/3rfm.pdb
```

To generate the CG protein model and structure, we will use the Python program `martinize2` ([Martinize2 and vermouth](#)). Its development is done on GitHub, and it is led by the Molecular Dynamics group at the University of Groningen. To prepare the protein structure, we first have to remove the water and additional molecules present. Often, detergents, ligands, and molecules added to facilitate the crystallization are present in experimental structures. Use the following command to clean the structure:

```
grep "^ATOM" 3rfm.pdb > 3rfm_clean.pdb
```

The protein structure contains 8 mutations (A54L, T88A, R107A, K122A, L202A, L235A, V239A, S277A) compared to the wild type (Doré *et al.*, 2011), which have to be reverted before starting the simulations. In addition, the extracellular loop 2 (ECL2) is missing in the structure 3RFM. One option to fix the mutations and the missing loop is to use the program MODELLER (57). Alternatively, we could rely on another experimental X-ray crystal structure, which contains the correct wild-type sequence and ECL2 loop—the `pdb` structure 2YDO (Lebon *et al.*, 2011). Currently, an alternative solution for fixing mutations and missing loops is simply to predict the A2A receptor structure based on the correct sequence using AlphaFold (Jumper *et al.*, 2021).

As a next step, we check the protonation state of the acidic and basic amino acids at pH 7.4 using the H++ server (<http://newbiophysics.cs.vt.edu/H++/>) (Anandakrishnan *et al.*, 2012). We follow the predictions of the protonation state for all amino acids except Asp52. Because it is quite deeply buried in the protein and, moreover, experiments showed that there is a Na⁺ ion in close proximity, we keep Asp52 neutral to compensate for the nearby counter ion which neutralizes the charge.

1.3.2 Generating the Martini 3 protein model

With the complete atomistic protein structure at hand, we can now generate the Martini 3 protein model. The program `martinize2` (Martinize2 and vermouth) converts the atomistic structure to a CG one, generates a protein model, and an overall system topology. This information is stored in the file format needed for the simulations with the program package GROMACS.

1.3.2.1 The elastic network model

The first option described here to maintain the higher-order protein structure is the EN model. It can be generated with `martinize2` using the option `-elastic`. In the EN model, harmonic bonds are added between backbone (BB) beads, which are in a certain distance range in the reference structure and have a sequence distance of ≥ 3 . BB beads with a shorter sequence distance are already interacting via bonds or angles. Also, EN bonds are added only between amino acids that belong to the same chain. The distance range for the EN can be specified with a lower (`-el`) and upper distance cutoff (`-eu`). They are usually chosen between 0.3–0.5 nm and 0.8–1.0 nm, respectively (Periole *et al.*, 2009). Typically, a fixed force constant (`-ef`) of 500–1000 kJ mol⁻¹ nm⁻² is used for the harmonic bonds, but the force constant can also be chosen to be distance-dependent. To optimize the EN, the force constant as well as the distance cutoffs can be adapted in order to improve the comparison with atomistic simulations or experimental data (Periole *et al.*, 2009).

Other options for the protein model which we need to specify are `-scfix` and `-cys auto`. The `-scfix` option adds side chain corrections based on the atomistic reference structure, which improves the side chain orientation. The option `-cys auto` enables `martinize2` to automatically detect and add disulfide bridges. In addition, the path to the program `dssp` (<https://github.com/cmbi/dssp>)

(Kabsch and Sander, 1983; and Touw *et al.*, 2015) is required, which is used to determine the secondary structure of the protein.

```
martinize2 -f atomistic_complete.pdb -o A2AR_only.top -x
A2AR_cg.pdb -dssp /path/to/dssp -p backbone -ff martini3001
-elastic -ef 500.0 -el 0.5 -eu 0.8 -scfix -cy auto
```

In the case of the structure 3RFM, the mutations could be reversed with the additional flag `-mutate` of `martinize2`. For instance, the mutation A54L could be reverted with the flag `-mutate A-LEU54:ALA`. We get three new files from `martinize2`, namely `A2AR_cg.pdb` containing the CG structure, `molecule_0.itp` containing the protein model, and a rudimentary master topology file `A2AR_only.top`. The latter has to be modified before starting a simulation of the A2A receptor embedded in a lipid bilayer; details are given in Sec. 1.1.4. As changes in the protonation states are still not part of `martinize2`, you may need to manually change the protonation of residues in the `molecule_0.itp` file. An example would be to change the protonation state of Asp52. For the protein model, this requires adjustments of bead 118 in the following line

```
118 SQ5n 52 ASP SC1 118 -1.0
```

by changing the bead type from SQ5n to SP2 and the charge from -1.0 to 0.0 .

```
118 SP2 52 ASP SC1 118 0.0 .
```

1.3.2.2 The Gō-like model

An alternative for the second layer of the Martini 3 protein model is to use a Gō-like model to maintain the higher-order protein structure. The combination of a Gō-like model with Martini—termed GōMartini—was introduced by Poma *et al.* (2017). Here, BB beads are connected on the basis of a contact map generated from an atomistic reference structure. Lennard–Jones potentials are used to describe the interactions between the connected BB beads. In an updated GōMartini version, virtual beads are placed at the BB bead position. They are connected by the Gō-like bonds implemented as regular non-bonded interactions, enabling them to benefit from the non-bonded cutoff scheme. The regular non-bonded interactions between the connected BB beads are excluded.

To generate a GōMartini model for your protein, you have to first use `martinize2`. The options `-govs-include` and `-govs-moltype` are required to generate an `itp` file as well as a CG structure.

```
martinize2 -f atomistic_complete.pdb -o A2AR_only.top -x
A2AR_cg.pdb -dssp /path/to/dssp -p backbone -ff martini3001
-govs-include -govs-moltype a2ar -scfix -cys auto
```

As described in the model with the elastic network, do not forget to revert possible mutations and define protonation states in your model. The next step is to obtain the contact map from the web-server <http://info.ifpan.edu.pl/~rcsu/rcsu/index.html> using the default setting for the radii and Fibonacci number (Wolek *et al.*, 2015). As an alternative, the user can also use Gō Contact Map web-server (<http://pomalab.ippt.pan.pl/GoContactMap/>). Please note that the requirements for the `pdb` file

uploaded to the web-server are quite strict. Therefore, please check carefully if your contact map is meaningful before using the contact map in the next step. In particular, the table listing the “Residue residue contacts” will be used.

For the next step, the script `create_goVirt.py` is required, which can be downloaded from <http://cgmartini.nl>. It requires the CG structure of the protein in the `pdb` format (`-s`), the number of CG beads in the protein excluding the virtual Gō beads (`--Natoms`), the number of missing residues in the atomistic structure (`--missres`), as well as the contact map file (`-f`). The prefix of the generated files can be specified (`--moltype`) to distinguish different protein chains which are not connected by Gō-like bonds. Note that if the Gō-like model will be combined with a CG protein generated by `martinize2`, the same name must be specified as for the `-govs-moltype` flag of `martinize2`. In addition, the dissociation energy of the Lennard-Jones potentials (`--go_eps`) as well as a lower (`--cutoff_short`) and upper cutoff (`--cutoff_long`) for the distance between two connected backbone beads have to be specified. The latter are similar to the EN model.

```
./create_goVirt.py -s A2AR_cg.pdb -f 3RFM_complete.map --moltype
a2ar --go_eps 12.0 --cutoff_short 0.3 --cutoff_long 1.1 --Natoms
717
```

With the above command, `create_goVirt.py` produces four files beginning with `a2ar_`, which contain details of the GōMartini model:

<code>a2ar_BB-part-def_VirtGoSites.itp</code>	bead definitions of the virtual particles
<code>a2ar_go-table_VirtGoSites.itp</code>	interaction table of the Gō-like bonds
<code>a2ar_exclusions_VirtGoSites.itp</code>	exclusions
<code>a2ar_go4view_harm.itp</code>	representing the Gō-like bonds for visualization

It is important not to change the names of the files because they are included in several other files requiring the correct file names. Finally, two `#include` statements have to be added to the `martini_v3.0.0.itp` file. Note that the following two commands should be executed only once.

```
sed -i "s/\[ nonbond_params\]/\#ifdef GO_VIRT\n\#include \"BB-part-
def_VirtGoSites.itp\"\n\#endif\n\n\ [ nonbond_params \]/"
martini_v3.0.0.itp
echo -e "\n\#ifdef GO_VIRT\n\#include \"go-
table_VirtGoSites.itp\"\n\#endif" >> martini_v3.0.0.itp
```

To activate the two `#include` statements, you need to define the variable `GO_VIRT` at the beginning of the master topology:

```
#define GO_VIRT
```

Make sure that it is included if you use the GōMartini model. If you use the Martini 3 force field without the GōMartini model in the future, you simply do not define `GO_VIRT` and the `#include` statements will be skipped.

16 July 2024 15:18:01

1.3.2.3 Fine-tuning the binding pocket of the Martini 3 model

The core as well as the binding pocket of the A2A receptor are quite hydrated (Liu *et al.*, 2012). To achieve a realistic behavior of the protein model, the hydration of the binding pocket as well as its hydrogen bonding network has to be taken into account during the model preparation.

A regular water bead W representing four water molecules is too large to be used in the binding pocket. Tiny water beads TW, which represent two water molecules, are much better suited for this task. To mimic the hydration of the binding pocket, place a TW bead in the binding pocket. Add the additional bead at the end of the coordinate file `A2AR_cg.pdb`. Use the coordinates of the SC1 bead of Asn181 as template and modify them slightly for instance by adding 0.1 nm. In the `molecule_0.itp` file, add a line at the end of the `[atoms]` section reading

```
718 TW    306 TW    TW    718    0.0
```

in the case of the EN model. For the GōMartini model, the atom number has to be 1023 instead of 718. To prevent the TW bead from diffusing out of the pocket, add two harmonic bonds to the side chains of Asn181 and His250 with a force constant of 500 kJ/(mol · nm²) and an equilibrium distance of 0.28 nm. Since the water bead is not covalently bound to the two amino acids, specify bond type 6, which does not exclude the non-bonded interactions. The additional lines in the `[bonds]` section of the `molecule_0.itp` file with the EN model read

```
410 718 6 0.28 500.0
571 718 6 0.28 500.0
```

To further stabilize the binding pocket, we add two harmonic bonds mimicking the water-mediated hydrogen bonds Thr88-Trp246 and Asn181-His250 (Liu *et al.*, 2012) again using bond type 6:

```
197 562 6 0.39 500.0
410 571 6 0.40 500.0
```

1.4 PROTEIN-LIGAND BINDING WITH CONVENTIONAL SIMULATIONS

Since we now have both the coordinates and topology of the Martini 3 protein, we can put the protein in a simulation cell and add lipids, water, and ions. We will use for this task insane (Wassenaar *et al.*, 2015)

```
python2 insane.py -f A2AR_cg.pdb -o box.gro -p CG.top -pbc cubic
-box 12,12,12 -u POPC -l POPC -salt 0.15 -sol W -d 0
```

Be aware that the current implementation of insane is still based on Python 2. The output file (`CG.gro`) contains NA+, CL-, and W residues, while `CG.top` is a prototype of the master topology file.

You can remove the charge signs of NA and CL in the `CG.top` file. You should also replace “Protein” with `molecule_0`. You may get a list of molecules which is similar to

```
[ molecules ]
; name      number
molecule_0 1
POPC        205
POPC        206
W           8800
NA          90
CL          101
```

Don't forget to also include the Martini 3 force field files at the beginning of the `CG.top`.

```
#include "martini_v3.0.0.itp"
#include "molecule_0.itp"
#include "martini_v3.0.0_phospholipids_v1.itp"
#include "martini_v3.0.0_solvents_v1.itp"
#include "martini_v3.0.0_ions_v1.itp"
```

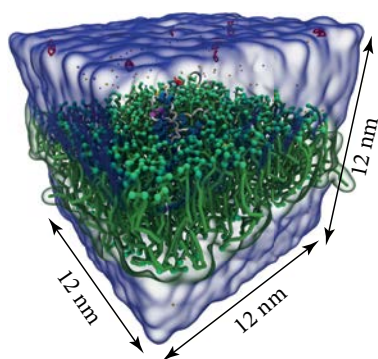


FIG. 1.4

Simulation box containing the A2A receptor embedded in a POPC bilayer (green) and ligands (red) solvated in water (transparent blue surface). Adapted from Ref. Souza *et al.*, Nat. Commun. 11, 3714 (2020) under CC-BY.

The first `itp` file contains information about non-bonded interactions between all types of Martini 3 beads. The other files are more specific and define the protein (`molecule_0.itp`, generated in Sec. 1.1.3) as well as standard Martini 3 molecules such as lipids, solvents, and ions. The last step is to add caffeine molecules to your simulation box. This can be done with the GROMACS tool `gmx insert-molecules`

```
gmx insert-molecules -f box.gro -nmol 10 -ci
caffeine_CG.gro -o box_ready.gro
```

We added 10 molecules of caffeine via the flag `-nmol`. This corresponds to a concentration of $16 \cdot 10^{-3}$ M. A representation of the simulation box is shown in Fig. 1.4.

At this point, almost all the steps for building your protein–ligand binding system are complete. You only need to add the caffeine `itp` file and the number of caffeine molecules in the section `[molecules]` in the topology file. We are now ready to perform the MD simulations and start studying the binding of caffeine to the A2A receptor. We suggest a minimum of 200 μ s of sampling to be able to observe multiple binding events. This could efficiently be performed with 10 replicas of 20 μ s.

Do you see any binding events taking place? Does caffeine bind directly to the A2A receptor or does it first go to the membrane? In case you did not observe binding events, you may consider increasing the total simulation time or using more simulation replicas. Take a look at our published study for more details and to get more ideas for analysis you could perform on your trajectory (Souza *et al.*, 2020).

1.5 OUTLOOK

Considering the constant efforts in improving the Martini force field and the new possible applications involving protein–ligand binding and virtual drug design (Souza *et al.*, 2020, 2021b), it seems critical to develop further the model and its associated tools in certain key directions, which are summarized below:

Ligand database and automated parametrization: One of the main limitations of the use of Martini in protein–ligand binding studies is the generation of accurate models for the ligands. Although there are well-defined protocols to generate these models (as described in this chapter), only automatic protocols will allow virtual screening of hundreds to millions of ligands, something that is already common in many *in silico* drug-design campaigns based on other models. The first step towards a fully automatic topology builder (ATB) in Martini 3 is the creation of a curated dataset of high-quality Martini 3 models. This effort is essential for the full calibration and benchmarks of any automatic protocol strategy. An initial set of this database was recently published (Alessandri *et al.*, 2022), while MAD (Martini Database—<https://mad.ibcp.fr/explore>) (Hilpert *et al.*, 2022) will become the standard platform to collect new models in the future. Although some ATBs have already been developed, more work is necessary, in particular to adapt and integrate these codes for Martini 3, which has an expanded set of beads covering a larger chemical space than the previous versions. Some of the already published ATB tools focus on generating bonded potentials from atomistic trajectories, such as Swarm-CG (Empereur-Mot *et al.*, 2020) and PyCGtool (Graham *et al.*, 2017). Bartender also fits in this category, but using a QM-MD strategy, which can be potentially advantageous for the parameterization of ligands not typically included (or accurate) in the chemical space of typical AA force fields. Other codes, like graph-based coarse-graining (Webb *et al.*, 2019), cg_param (Potter *et al.*, 2021), and autoMartini (Bereau and Kremer, 2015) addressed the challenge of bead assignment and mapping of a chemical structure to its (Martini 2) CG representation. Future developments should target a fully integrated code converting a SMILES string into an accurate Martini 3 model in terms of mapping, bead assignment, and bonded parameters. Additionally, similar protocols for automatic parameterization of titratable Martini models (Grünewald *et al.*, 2020) should help with the challenge of capturing protonation changes in ligand binding to proteins.

Protein conformational flexibility: Although the representation of protein flexibility possibly improved with the combination of Martini 3 and Gō-like models (Poma *et al.*, 2017; and Souza *et al.*, 2019), there are still some important limitations blocking its full potential, in particular for certain problems

involving protein-small molecule binding. While developing a Martini model capable of folding proteins from their sequence could be considered too ambitious, problems such as the discovery of cryptic pockets may be within the reach of the model. For instance, the current approach to generate contact maps of GōMartini is based on a unique reference structure. New developments using atomistic trajectories could result in higher quality contact maps and refinements in the depth of the Gō LJ potentials. Side chain dihedrals are currently added based on a reference folded structure [via side chain fixes ([Herzog et al., 2016](#))], but new improvements could also make them more generic and useful to describe conformational transitions. Additionally, a more realistic description of the backbone, such as an explicit representation of peptide-group dipoles, would already be a significant improvement. For instance, modeling the realistic transition between secondary structure elements would be extremely beneficial. Such an idea has been used in other coarse-grained models for proteins (e.g., [Sieradzan et al., 2022](#)), and it has already been proposed to be incorporated in Martini ([Marrink and Tieleman, 2013](#)). Overall, we foresee many possible directions to improve Martini 3 protein models in the near future.

Generic backmapping codes: Even with improvements in the generation of accurate protein and ligand models, CG approaches such as Martini have some inherent limitations (see Note 2). Possibly any virtual ligand-screening strategy using Martini will need a refinement step based on atomistic models ([Souza et al., 2021a](#)), which fully depends on accurate protocols to reverse transform CG configurations into fine-grained structures at united-atom and all-atom resolutions. Different options of backmapping already exist, such as fragment-based [e.g., CG2AT ([Stansfeld and Sansom, 2011](#); and [Vickery and Stansfeld, 2021](#))], machine learning-based [e.g., GLIMPS ([Louison et al., 2021](#))], and geometric approaches [e.g., backward ([Wassenaar et al., 2014](#))]. Any of these strategies are currently limited to only a set of molecules, with the main challenge still not solved: a general tool to automatically transform any molecule between atomistic and CG resolutions, generating both coordinate and topology files. Such a development would be appreciated not only in protein–ligand binding studies but also for the overall modeling community using Martini models.

1.6 NOTES

1. *Computational performance of Martini:* The low cost of MD simulations with Martini comes from a combination of factors. First of all, given the resolution of the beads, it is expected that a simulation box with Martini will have 4 to 10 times fewer particles than all-atom systems. Given the higher masses of the beads in combination with the use of constraints and virtual particles (typically employed in systems such as aromatic rings), Martini can run simulations with a time step of 20 fs. This time step gives a speed-up of 1 order of magnitude in relation to atomistic models, which instead typically require a time step of 2 fs. Additional performance gains come from the use of shifted Coulomb potentials, computationally cheaper than PME. A final important aspect to be considered is the effective timescale in CG simulations. The CG dynamics are faster than the

AA dynamics because the CG potential landscape is much smoother as compared to an atomistic one. Estimates with neutral small molecules in Martini 3 indicate that diffusion coefficients are 2–3 times higher than experimental references (Souza *et al.*, 2020). These values can be larger in highly charged systems such as ionic liquids (Vazquez-Salazar *et al.*, 2020), deep eutectic solvents (Vainikka *et al.*, 2021), and PIP-containing membrane (Borges-Araújo *et al.*, 2022), given that the electrostatic interactions are underestimated in the non-polarizable Martini model. The combination of all of these factors indicates that Martini can reach speed-ups of 2–3 orders of magnitude as compared to all-atom simulations. Applications involving virtual screening could potentially get extra speed-up with coarse-graining in the chemical space, as recently demonstrated by Kanekal and Bureau (2019).

2. *General limitations of coarse-grained models:* The process of coarse-graining creates a number of limitations for these models, which are also the case for Martini 3 (Marrink and Tieleman, 2013; and Jarin *et al.*, 2021). An important example is the entropy loss upon coarse-graining, which is necessarily compensated for by a reduced enthalpy term (Baron *et al.*, 2007). As a consequence, the enthalpy/entropy balance of many processes can be biased when modeled at the CG level, which can also affect its temperature dependence. Another limitation is the lack of directionality of certain intermolecular interactions, such as hydrogen bonding or T-shaped interactions between aromatic compounds (Alessandri *et al.*, 2022). This issue may be one of the reasons hindering the development of a foldable protein backbone model in Martini. An additional problem of CG models is the non-trivial interpretation of the timescale, which is caused by the non-homogeneously smoothed potential energy surface on which the CG dynamics take place (Marrink and Tieleman, 2013; and Jarin *et al.*, 2021). Although for certain systems, the speed-up of Martini models in relation to experimental diffusion rates was quantified (Souza *et al.*, 2020; Vazquez-Salazar *et al.*, 2020; and Borges-Araújo *et al.*, 2022), the interpretation of timescales should always be taken with caution. The Martini 3 publication presents additional information in this matter for the reader, in particular in the discussion section and supplementary information (Souza *et al.*, 2021a).
3. *The stickiness problem in Martini 2:* Despite the success of the Martini force field, in the recent years, it became clear that the Martini CG approach had some inherent problems affecting the model. Several studies reported an apparent overstabilization of biomolecular interactions. In particular, high aggregation propensity was reported for soluble (Stark *et al.*, 2013) and membrane proteins (Javanainen *et al.*, 2017; and Majumder and Straub, 2021), as well as carbohydrates (Gu *et al.*, 2017; Schmalhorst *et al.*, 2017; and Shivgan *et al.*, 2020). Known as the *stickiness problem*, many *ad hoc* solutions were proposed, including scaling of solute-solute or solute-solvent interactions. The usage of S-beads for the modeling of carbohydrate rings was also a strategy used in the re-parameterization of gangliosides (Gu *et al.*, 2017). The underlying cause of the stickiness problem was finally elucidated in the work of Alessandri *et al.* (2019), showing that the problem was related to an imbalance between the small and regular bead types, in combination with the use of weak bonded force constants or bonds between beads that were too short in the

description of biomolecules (Alessandri *et al.*, 2019). New rules for the use of different bead sizes and for the calculation of bond lengths together with the full re-parameterization of the model were part of the main improvements implemented in Martini 3 (Souza *et al.*, 2021a) to tackle the stickiness problem. Despite the fact that Martini 3 seems to show an overall improvement for the interactions between folded proteins and peptides (Lamprakis *et al.*, 2021; and Souza *et al.*, 2021a) and carbohydrates (Grünwald *et al.*, 2022), some problems may persist in the case of intrinsically disordered proteins (Thomasen *et al.*, 2022).

4. *Labels in Martini 3*: Specific subtypes, called “labels,” are an alternative and complementary solution to fine-tune CG models in Martini 3. The concept already existed in Martini 2, but it only considered the different hydrogen bonding capabilities of N and Q beads, keeping these beads with a similar degree of hydrophobicity. In Martini 3, there are many more label options that can be applied to all beads, slightly changing their behavior, which is translated into small changes in the interaction matrix. Hydrogen bonding, electron polarizability, positive/negative charge labels are chemically specific, which means that they are only applied to certain chemical types. Self-interaction and partial charge labels are more generic and can be applied on top of all beads of the organic block, including beads that have already been modified with some of the chemically specific labels. Labels can be used to promote selectivity in the interactions which are enhanced for smaller beads. For instance, hydrogen donor (“d”) and acceptor (“a”) increase by one interaction level the interaction of SP4d–SP4a in relation to SP4–SP4. On the other hand, TP4d–TP4a increased the TP4–TP4 interaction by three interaction levels. More details about labels are described in Sec. A4 of the supporting information of the Martini 3 publication (Souza *et al.*, 2021a).
5. *Mapping of branched fragments*: Fully branched fragments (such as fragments with quaternary carbon atoms or tertiary amine groups) should usually use beads of smaller size as the central atom is not exposed to the environment which reduces its influence on the interactions. For instance, a neopentane group contains 5 non-hydrogen atoms but as it is fully branched can be safely modeled as an R-bead. The choice of the bead size for partially branched fragments of molecules is not straightforward; it is desirable to validate the choice by comparison to experimental data and/or in relation to other Martini 3 CG models. The main danger of using the “fully branching rule” for partially branched fragments is to reduce the self-interactions too much. Such a reduction can be especially problematic in polymers where a small deviation on the monomer scale may add up to a big deviation at the polymer scale.
6. *COG-mapping of an atomistic trajectory*: If one wants to use the GROMACS tool `gmx traj` to map the AA trajectory to CG resolution using the COG strategy [alternatives include, for example, using Python and the MDAnalysis library (Gowers *et al.*, 2016)], a little workaround is needed. The `gmx traj` tool only allows center-of-mass (COM) mapping. Hence, we first need to create an AA GROMACS `topr` file with all the atoms of the atomistic structure having *the same mass*. This is done by modifying the original topology file obtained from the LigParGen server (CAFF_LigParGen.itp) by changing the values under the eighth column of the `[atoms]` section to an equal value (e.g., 1.0). Let us call the new file so created `CAFF_LigParGen_COG.itp`. It is this file that can be

used to create the GROMACS tpr file for the subsequent `gmx traj` command, where the `-com` flag of the `gmx traj` command will effectively act as a `-cog` flag.

7. *Exceptions in bead assignment:* In our experience, mapping of some functional groups, especially when done at higher resolutions (e.g., in aromatic rings), can vary based on the *proximity* of other functional groups. The rule of thumb is that such perturbations may require a ± 1 shift in the degree of polarity of the bead in question with respect to the “standard” choice listed in the reference “building block table” (Martini3).
8. *Mapping based on experience:* Once again, when creating a Martini 3 model for a new small molecule, it is useful to consult *already-developed models* in order to develop models in a rapid and more consistent way. In particular for small molecules, many examples are available on the Martini 3 small molecule GitHub repo (Alessandri *et al.*, 2022).
9. *Additional validation targets:* In general, besides free energies of transfer, other validation targets for the small molecule CG model may be considered depending on the application. In the case of protein–ligand interactions, as described, *the validation of the molecular size and shape* is also strongly encouraged. Together with the free energy of transfer comparison, it allows for fine-tuning and optimizing bead type choices and bonded parameters.

ACKNOWLEDGMENTS

The authors would like to thank all researchers that contributed to the development of the new Martini 3 force field. Additionally, we would like to acknowledge all users of the molecular modeling community that tested the open-beta version of Martini 3 and sent us feedback. Without your input and criticism, it would not have been possible to finish this new version of Martini.

REFERENCES

- Abellón-Ruiz, J., Kaptan, S. S., Baslé, A., Claudi, B., Bumann, D., Kleinekathöfer, U., and Van Den Berg, B., “Structural basis for maintenance of bacterial outer membrane lipid asymmetry,” *Nat. Microbiol.* **2**, 1616–1623 (2017).
- Abraham, M. J., Murtola, T., Schulz, R., Páll, S., Smith, J. C., Hess, B., and Lindahl, E., “GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers,” *SoftwareX* **1–2**, 19–25 (2015).
- Agostino, M. D., Risselada, H. J., Lürick, A., Ungermann, C., and Mayer, A., “A tethering complex drives the terminal stage of SNARE-dependent membrane fusion,” *Nature* **551**, 634–638 (2017).
- Alessandri, R., Barnoud, J., Gertsen, A. S. *et al.*, “Martini 3 coarse-grained force field: Small molecules,” *Adv. Theory Simulations* **5**, 2100391 (2022).
- Alessandri, R., Grünewald, F., and Marrink, S. J., “The martini model in materials science,” *Adv. Mater.* **33**, 2008635 (2021).

- Alessandri, R., Souza, P. C. T., Thallmair, S., Melo, M. N., de Vries, A. H., and Marrink, S. J., "Pitfalls of the martini model," *J. Chem. Theory Comput.* **15**, 5448–5460 (2019).
- Anandakrishnan, R., Aguilar, B., and Onufriev, A. V., "H++ 3.0: Automating pK prediction and the preparation of biomolecular structures for atomistic molecular modeling and simulations," *Nucleic Acids Res.* **40**, W537–W541 (2012).
- Atsmon-Raz, Y. and Tieleman, D. P., "Parameterization of palmitoylated cysteine, farnesylated cysteine, geranylgeranylated cysteine, and myristoylated glycine for the martini force field," *J. Phys. Chem. B* **121**, 11132–11143 (2017).
- Bannwarth, C., Caldeweyher, E., Ehlert, S., Hansen, A., Pracht, P., Seibert, J., Spicher, S., and Grimme, S., "Extended tight-binding quantum chemistry methods," *Wiley Interdiscip. Rev. Comput. Mol. Sci.* **11**, e1493 (2021).
- Baron, R., Trzesniak, D., de Vries, A. H., Elsener, A., Marrink, S. J., and van Gunsteren, W. F., "Comparison of thermodynamic properties of coarse-grained and atomic-level simulation models," *ChemPhysChem* **8**, 452–461 (2007).
- Bartender, see <https://github.com/rmera/bartender>.
- Bereau, T. and Kremer, K., "Automated parametrization of the coarse-grained martini force field for small organic molecules," *J. Chem. Theory Comput.* **11**, 2783–2791 (2015).
- Borges-Araújo, L., Souza, P. C. T., Fernandes, F., and Melo, M. N., "Improved parameterization of phosphatidylinositide lipid headgroups for the Martini 3 coarse-grain force field," *J. Chem. Theory Comput.* **18**, 357–373 (2022).
- Bruininks, B. M. H., Souza, P. C. T., Ingólfsson, H., and Marrink, S. J., "A molecular view on the escape of lipoplexed DNA from the endosome," *Elife* **9**, e52012 (2020).
- CGbuilder, "CGbuilder: Build coarse-grained mapping for molecules from a web-GUI," see <https://github.com/jbarnoud/cgbuilder>.
- De Jong, D. H., Baoukina, S., Ingólfsson, H. I., and Marrink, S. J., "Martini straight: Boosting performance using a shorter cutoff and GPUs," *Comput. Phys. Commun.* **199**, 1–7 (2016).
- De Jong, D. H., Liguori, N., Van Den Berg, T., Arnarez, C., Periole, X., and Marrink, S. J., "Atomistic and coarse grain topologies for the cofactors associated with the photosystem II core complex," *J. Phys. Chem. B* **119**, 7791–7803 (2015).
- De Jong, D. H., Singh, G., Bennett, W. F. D., Arnarez, C., Wassenaar, T. A., Schäfer L. V., Periole, X., Tieleman, D. P., and Marrink, S. J., "Improved parameters for the martini coarse-grained protein force field," *J. Chem. Theory Comput.* **9**, 687–697 (2013).
- De Marco, A. L., Bochicchio, D., Gardin, A., Doni, G., and Pavan, G. M., "Controlling exchange pathways in dynamic supramolecular polymers by controlling defects," *ACS Nano* **15**, 14229–14241 (2021).
- Dodda, L. S., De Vaca, I. C., Tirado-Rives, J., and Jorgensen, W. L., "Ligpargen web server: An automatic OPLS-AA parameter generator for organic ligands," *Nucleic Acids Res.* **45**, W331–W336 (2017a).
- Dodda, L. S., Vilseck, J. Z., Tirado-Rives, J., and Jorgensen, W. L., "1.14*CM1A-LBCC: Localized bond-charge corrected CM1A charges for condensed-phase simulations," *J. Phys. Chem. B* **121**, 3864–3870 (2017b).

- Doré, A. S., Robertson, N., Errey, J. C. *et al.*, “Structure of the adenosine A2A receptor in complex with ZM241385 and the xanthines XAC and caffeine,” *Structure* **19**, 1283–1293 (2011).
- Eastman, P., Swails, J., Chodera, J. D. *et al.*, “OpenMM 7: Rapid development of high performance algorithms for molecular dynamics,” *PLOS Comput. Biol.* **13**, e1005659 (2017).
- Empereur-Mot, C., Pesce, L., Doni, G., Bochicchio, D., Capelli, R., Perego, C., and Pavan, G. M., “Swarm-CG: Automatic parametrization of bonded terms in MARTINI-based coarse-grained models of simple to complex molecules via fuzzy self-tuning particle swarm optimization,” *ACS Omega* **5**, 32823–32843 (2020).
- Faustino, I., Abdizadeh, H., Souza, P. C. T., Jeucken, A., Stanek, W. K., Guskov, A., Slotboom, D. J., and Marrink, S. J., “Membrane mediated toppling mechanism of the folate energy coupling factor transporter,” *Nat. Commun.* **11**, 1763 (2020).
- Gowers, R. J., Linke, M., Barnoud, J., Reddy, T. J. E., Melo, M. N., Seyler, S. L., Dotson, D. L., Domanski, J., Buchoux, S., Kenney, I. M., and Beckstein, O., “MDAnalysis: A Python package for the rapid analysis of molecular dynamics simulations,” in *Proceedings of the 15th Python in Science Conference*, edited by S. Benthall and S. Rostrup, Austin, TX, 2016. (U.S. Department of Energy, 2019), pp. 98–105
- Graham, J. A., Essex, J. W., and Khalid, S., “PyCGTOOL: Automated generation of coarse-grained molecular dynamics models from atomistic trajectories,” *J. Chem. Inf. Model* **57**, 650–656 (2017).
- GROMACS 2021.5, “GROMACS 2021.5 documentation: File formats,” see <https://manual.gromacs.org/documentation/current/reference-manual/topologies/topology-file-formats.html>.
- Grünewald, F., Punt, M. H., Jefferys, E. E. *et al.*, “Martini 3 coarse-grained force field for carbohydrates,” *J. Chem. Theory Comput.* (in press) (2022).
- Grünewald, F., Souza, P. C. T., Abdizadeh, H., Barnoud, J., de Vries, A. H., and Marrink, S. J., “Titratable martini model for constant pH simulations,” *J. Chem. Phys.* **153**, 024118 (2020).
- Gu, R. X., Ingólfsson, H. I., de Vries, A. H., Marrink, S. J., and Tieleman, D. P., “Ganglioside-lipid and ganglioside-protein interactions revealed by coarse-grained and atomistic molecular dynamics simulations,” *J. Phys. Chem. B* **121**, 3262–3275 (2017).
- Hansch, C., Leo, A., and Hoekman, D., *Exploring QSAR, Vol. 2: Hydrophobic, Electronic, and Steric Constants* (American Chemical Society, Washington, DC, 1995).
- Herzog, F. A., Braun, L., Schoen, I., and Vogel, V., “Improved side chain dynamics in MARTINI simulations of protein-lipid interfaces,” *J. Chem. Theory Comput.* **12**, 2446–2458 (2016).
- Hess, B., Bekker, H., Berendsen, H. J. C., and Fraaije, J. G. E. M., “LINCS: A linear constraint solver for molecular simulations,” *J. Comput. Chem.* **18**, 1463–1472 (1997).
- Hilpert, C., Beranger, L., Souza, P. C. T., Vainikka, P. A., Nieto, V., Marrink, S. J., Monticelli, L., and Launay, G., “Facilitating CG simulations with MAD: The MARTini Database Server,” *bioRxiv* (2022), see <https://doi.org/10.1101/2022.08.03.502585>.
- Humphrey, W., Dalke, A., and Schulten, K., “VMD—Visual molecular dynamics,” *J. Molec. Graph* **14**, 33–38 (1996).
- Jarin, Z., Newhouse, J., and Voth, G. A., “Coarse-grained force fields from the perspective of statistical mechanics: Better understanding of the origins of a MARTINI hangover,” *J. Chem. Theory Comput.* **17**, 1170–1180 (2021).

- Javanainen, M., Martinez-Seara, H., and Vattulainen, I., "Excessive aggregation of membrane proteins in the Martini model," *PLoS One* **12**, e0187936 (2017).
- Jewett, A. I., Stelter, D., Lambert, J. *et al.*, "Moltemplate: A tool for coarse-grained modeling of complex biological matter and soft condensed matter physics," *J. Mol. Biol.* **433**, 166841 (2021).
- Jorgensen, W. L., Chandrasekhar, J., Madura, J. D., Impey, R. W., and Klein, M. L., "Comparison of simple potential functions for simulating liquid water," *J. Chem. Phys.* **79**, 926 (1998).
- Jorgensen, W. L. and Tirado-Rives, J., "Potential energy functions for atomic-level simulations of water and organic and biomolecular systems," *Proc. Natl. Acad. Sci. U.S.A.* **102**, 6665–6670 (2005).
- Jumper, J., Evans, R., Pritzel, A. *et al.*, "Highly accurate protein structure prediction with AlphaFold," *Nature* **596**, 583–589 (2021).
- Kabsch, W. and Sander, C., "Dictionary of protein secondary structure: Pattern recognition of hydrogen-bonded and geometrical features," *Biopolymers* **22**, 2577–2637 (1983).
- Kanekal, K. H. and Berau, T., "Resolution limit of data-driven coarse-grained models spanning chemical space," *J. Chem. Phys.* **151**, 164106 (2019).
- Kmiecik, S., Gront, D., Kolinski, M., Wieteska, L., Dawid, A. E., and Kolinski, A., "Coarse-grained protein models and their applications," *Chem. Rev.* **116**, 7898–7936 (2016).
- Lamprakis, C., Andreadelis, I., Manchester, J., Velez-Vega, C., Duca, J. S., and Cournia, Z., "Evaluating the efficiency of the martini force field to study protein dimerization in aqueous and membrane environments," *J. Chem. Theory Comput.* **17**, 3088–3102 (2021).
- Lebon, G., Warne, T., Edwards, P. C., Bennett, K., Langmead, C. J., Leslie, A. G. W., and Tate, C. G., "Agonist-bound adenosine A2A receptor structures reveal common features of GPCR activation," *Nature* **474**, 521–525 (2011).
- Liaci, A. M., Steigenberger, B., Souza, P. C. T., Tamara, S., Gröllers-Mulderij, M., Ogrissek, P., Marrink, S. J., Scheltema, R. A., and Förster, F., "Structure of the human signal peptidase complex reveals the determinants for signal peptide cleavage," *Mol. Cell* **81**, 3934–3948 (2021).
- Liu, J., Qiu, L., Alessandri, R. *et al.*, "Enhancing molecular n-type doping of donor–acceptor copolymers by tailoring side chains," *Adv. Mater.* **30**, 1704630 (2018).
- Liu, W., Chun, E., Thompson, A. A. *et al.*, "Structural basis for allosteric regulation of GPCRS by sodium ions," *Science (80–)* **337**, 232–236 (2012).
- López, C. A., Bellesia, G., Redondo, A., Langan, P., Chundawat, S. P. S., Dale, B. E., Marrink, S. J., and Gnanakaran, S., "MARTINI coarse-grained model for crystalline cellulose microfibers," *J. Phys. Chem. B* **119**, 465–473 (2015).
- López, C. A., Rzepiela, A. J., de Vries, A. H., Dijkhuizen, L., Hünenberger, P. H., and Marrink, S. J., "Martini coarse-grained force field: Extension to carbohydrates," *J. Chem. Theory Comput.* **5**, 3195–3210 (2009).
- López, C. A., Sovova, Z., Van Eerden, F. J., De Vries, A. H., and Marrink, S. J., "Martini force field parameters for glycolipids," *J. Chem. Theory Comput.* **9**, 1694–1708 (2013).
- Louison, K. A., Dryden, I. L., and Laughton, C. A., "GLIMPS: A machine learning approach to resolution transformation for multiscale modeling," *J. Chem. Theory Comput.* **17**, 7930–7937 (2021).

- Machado, N., Bruininks, B. M. H., Singh, P. *et al.*, “Complex nanoemulsion for vitamin delivery: Droplet organization and interaction with skin membranes,” *Nanoscale* **14**, 506–514 (2022).
- Majumder, A. and Straub, J. E., “Addressing the excessive aggregation of membrane proteins in the MARTINI model,” *J. Chem. Theory Comput.* **17**, 2513–2521 (2021).
- Marrink, S. J., de Vries, A. H., and Mark, A. E., “Coarse grained model for semiquantitative lipid simulations,” *J. Phys. Chem. B* **108**, 750–760 (2004).
- Marrink, S. J. and Mark, A. E., “Molecular dynamics simulation of the formation, structure, and dynamics of small phospholipid vesicles,” *J. Am. Chem. Soc.* **125**, 15233–15242 (2003a).
- Marrink, S. J. and Mark, A. E., “The mechanism of vesicle fusion as revealed by molecular dynamics simulations,” *J. Am. Chem. Soc.* **125**, 11144–11145 (2003b).
- Marrink, S. J., Risselada, H. J., Yefimov, S., Tieleman, D. P., and de Vries, A. H., “The MARTINI force field: Coarse grained model for biomolecular simulations,” *J. Phys. Chem. B* **111**, 7812–7824 (2007).
- Marrink, S. J. and Tieleman, D. P., “Perspective on the Martini model,” *Chem. Soc. Rev.* **42**, 6801–6822 (2013).
- Martini3, “Martini3 small molecules: building block table,” see https://github.com/ricalessandri/Martini3-small-molecules/blob/main/tutorials/building_block_table.pdf.
- Martinize2 and vermouth, “Martinize2 and vermouth: The ultimate resolution transformation tools,” see <https://github.com/marrink-lab/vermouth-martinize> (last accessed January 19, 2022).
- Melo, M. N., Ingólfsson, H. I., and Marrink, S. J., “Parameters for Martini sterols and hopanoids based on a virtual-site description,” *J. Chem. Phys.* **143**, 243152 (2015).
- Monticelli, L., Kandasamy, S. K., Periole, X., Larson, R. G., Tieleman, D. P., and Marrink, S. J., “The MARTINI coarse-grained force field: Extension to proteins,” *J. Chem. Theory Comput.* **4**, 819–834 (2008).
- Periole, X., Cavalli, M., Marrink, S. J., and Ceruso, M. A., “Combining an elastic network with a coarse-grained molecular force field: Structure, dynamics, and intermolecular recognition,” *J. Chem. Theory Comput.* **5**, 2531–2543 (2009).
- Phillips, J. C., Hardy, D. J., Maia, J. D. C. *et al.*, “Scalable molecular dynamics on CPU and GPU architectures with NAMD,” *J. Chem. Phys.* **153**, 044130 (2020).
- Poma, A. B., Cieplak, M., and Theodorakis, P. E., “Combining the MARTINI and structure-based coarse-grained approaches for the molecular dynamics studies of conformational transitions in proteins,” *J. Chem. Theory Comput.* **13**, 1366–1374 (2017).
- Potter, T. D., Barrett, E. L., and Miller, M. A., “Automated coarse-grained mapping algorithm for the martini force field and benchmarks for membrane-water partitioning,” *J. Chem. Theory Comput.* **17**, 5791 (2021).
- Rowland, R. S. and Taylor, R., “Intermolecular nonbonded contact distances in organic crystal structures: Comparison with distances expected from van der Waals Radii,” *J. Phys. Chem.* **100**, 7384–7391 (1996).
- Schmalhorst, P. S., Deluweit, F., Scherrers, R., Heisenberg, C. P., and Sikora, M., “Overcoming the limitations of the MARTINI force field in simulations of polysaccharides,” *J. Chem. Theory Comput.* **13**, 5039–5053 (2017).

- Schrödinger, “The PyMOL Molecular Graphics System, Version 1.8,” (2015).
- Shivgan, A. T., Marzinek, J. K., Huber, R. G., Krah, A., Henchman, R. H., Matsudaira, P., Verma, C. S., and Bond, P. J., “Extending the martini coarse-grained force field to N-glycans,” *J. Chem. Inf. Model* **60**, 3864–3883 (2020).
- Sieradzan, A. K., Czaplewski, C., Krupa, P. *et al.*, “Modeling the structure, dynamics, and transformations of proteins with the UNRES force field—Methods in molecular biology,” in *Protein Fold*, edited by V. Muñoz (Humana, New York, NY, 2022), pp. 399–416.
- Sousa, F. M., Lima, L. M. P., Arnarez, C., Pereira, M. M., and Melo, M. N., “Coarse-grained parameterization of nucleotide cofactors and metabolites: Protonation constants, partition coefficients, and model topologies,” *J. Chem. Inf. Model* **61**, 335–346 (2021).
- Souza, P. C. T., Alessandri, R., Barnoud, J. *et al.*, “Martini 3: A general purpose force field for coarse-grained molecular dynamics,” *Nat. Methods* **18**, 382–388 (2021a).
- Souza, P. C. T., Limongelli, V., Wu, S., Marrink, S. J., and Monticelli, L., “Perspectives on high-throughput ligand/protein docking with martini MD simulations,” *Front. Mol. Biosci.* **8**, 199 (2021b).
- Souza, P. C. T., Thallmair, S., Conflitti, P., Ramírez-Palacios, C., Alessandri, R., Raniolo, S., Limongelli, V., and Marrink, S. J., “Protein–ligand binding with the coarse-grained Martini model,” *Nat. Commun.* **11**, 3714 (2020).
- Souza, P. C. T., Thallmair, S., Marrink, S. J., and Mera-Adasme, R., “An allosteric pathway in copper, zinc superoxide dismutase unravels the molecular mechanism of the G93A amyotrophic lateral sclerosis-linked mutation,” *J. Phys. Chem. Lett.* **10**, 7740–7744 (2019).
- Stansfeld, P. J. and Sansom, M. S. P., “From coarse grained to atomistic: A serial multiscale approach to membrane protein simulations,” *J. Chem. Theory Comput.* **7**, 1157–1166 (2011).
- Stark, A. C., Andrews, C. T., and Elcock, A. H., “Toward optimized potential functions for protein–protein interactions in aqueous solutions: Osmotic second virial coefficient calculations using the MARTINI coarse-grained force field,” *J. Chem. Theory Comput.* **9**, 4176–4185 (2013).
- Thomassen, F. E., Pesce, F., Roesgaard, M. A., Tesei, G., and Lindorff-Larsen, K., “Improving Martini 3 for disordered and multidomain proteins,” *J. Chem. Theory Comput.* **18**, 2033–2041 (2022).
- Touw, W. G., Baakman, C., Black, J., Te Beek, T. A. H., Krieger, E., Joosten, R. P., and Vriend, G., “A series of PDB-related databanks for everyday needs,” *Nucleic Acids Res.* **43**, D364–D368 (2015).
- Uusitalo, J. J., Ingólfsson, H. I., Akhshi, P., Tieleman, D. P., and Marrink, S. J., “Martini coarse-grained force field: Extension to DNA,” *J. Chem. Theory Comput.* **11**, 3932–3945 (2015).
- Uusitalo, J. J., Ingólfsson, H. I., Marrink, S. J., and Faustino, I., “Martini coarse-grained force field: Extension to RNA,” *Biophys. J.* **113**, 246–256 (2017).
- Vainikka, P., Thallmair, S., Souza, P. C. T., and Marrink, S. J., “Martini 3 coarse-grained model for type III deep eutectic solvents: Thermodynamic, structural, and extraction properties,” *ACS Sustain. Chem. Eng.* **9**, 17338–17350 (2021).
- Vazquez-Salazar, L. I., Selle, M., de Vries, A. H., Marrink, S. J., and Souza, P. C. T., “Martini coarse-grained models of imidazolium-based ionic liquids: From nanostructural organization to liquid–liquid extraction,” *Green Chem.* **22**, 7376–7386 (2020).

- Vickery, O. N. and Stansfeld, P. J., “CG2AT2: An enhanced fragment-based approach for serial multi-scale molecular dynamics simulations,” *J. Chem. Theory Comput.* **17**, 6472–6482 (2021).
- Vögele, M., Köfinger, J., and Hummer, G., “Hydrodynamics of diffusion in lipid membrane simulations,” *Phys. Rev. Lett.* **120**, 268104 (2018).
- Wassenaar, T. A., Ingólfsson, H. I., Böckmann, R. A., Tieleman, D. P., and Marrink, S. J., “Computational lipidomics with insane: A versatile tool for generating custom membranes for molecular simulations,” *J. Chem. Theory Comput.* **11**, 2144–2155 (2015).
- Wassenaar, T. A., Pluhackova, K., Böckmann, R. A., Marrink, S. J., and Tieleman, D. P., “Going backward: A flexible geometric approach to reverse transformation from coarse grained to atomistic models,” *J. Chem. Theory Comput.* **10**, 676–690 (2014).
- Webb, M. A., Delannoy, J. Y., and De Pablo, J. J., “Graph-based approach to systematic molecular coarse-graining,” *J. Chem. Theory Comput.* **15**, 1199–1208 (2019).
- Wolek, K., Gómez-Sicilia, À., and Cieplak, M., “Determination of contact maps in proteins: A combination of structural and chemical approaches,” *J. Chem. Phys.* **143**, 243105 (2015).
- Yen, H.-Y., Hoi, K. K., Liko, I. *et al.*, “Ptdins(4,5)P2 stabilizes active states of GPCRs and enhances selectivity of G-protein coupling,” *Nature* **559**, 423–427 (2018).
- Zhang, S., Huang, G., Versloot, R. C. A., Bruininks, B. M. H., Souza, P. C. T., Marrink, S. J., and Maglia, G., “Bottom-up fabrication of a proteasome–nanopore that unravels and processes single proteins,” *Nat. Chem.* **13**, 1192–1199 (2021).
- Zhang, X., Sundram, S., Oppelstrup, T., Korkila-Schumacher, S. I. L., Carpenter, T. S., Ingólfsson, H. I., Streitz, F. H., Lightstone, F. C., and Glosli, J. N., “ddcMD: A fully GPU-accelerated molecular dynamics program for the Martini force field,” *J. Chem. Phys.* **153**, 045103 (2020).